

内部言語の健全性定理と完全性定理

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概要

トポスの内部言語における健全性定理と完全性定理を証明する。

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$\mathcal{E}$ をトポス、 $\mathfrak{L}(\mathcal{E})$ を内部言語、 $\mathcal{M}(\mathcal{E})$ を標準モデル、 X を変数の有限集合とする。

文脈を増やすときは $\llbracket U \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})}$ というように、添字に変数を追加して書く。これは $\llbracket U \rrbracket_{X \cup \{y\}}^{\mathcal{M}(\mathcal{E})}$ と同じ意味。

1. 健全性定理

1.1. 補題たち

► 補題. 1.1 (文脈除去補題). $X \vdash U : \tau, x : \sigma, x \notin X$ ならば

$$\llbracket U \rrbracket_{X \cup \{x\}}^{\mathcal{M}(\mathcal{E})} = \llbracket U \rrbracket_X^{\mathcal{M}(\mathcal{E})} \circ \pi_{X \mathcal{M}(\mathcal{E})} : X^{\mathcal{M}(\mathcal{E})} \times \sigma^{\mathcal{M}(\mathcal{E})} \rightarrow X^{\mathcal{M}(\mathcal{E})} \rightarrow \tau^{\mathcal{M}(\mathcal{E})}$$

(文脈は適当に除去できる)

► 補題. 1.2 (代入補題). 項 $X \cup \{y : \tau\} \vdash U : \sigma, X \vdash V : \tau, y \notin X$ に対し

$$\llbracket U[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})} = \llbracket U \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1_{X \mathcal{M}(\mathcal{E})}, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})})$$

(右辺は当に代入)

【証明】 項 U の構造に関して再帰的に証明する。

1. U が変数 $U = x : \sigma$ の時、

(a) $x = y$ なら $\sigma = \tau$ であって

$$\llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})} = \llbracket x[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})} = (X^{\mathcal{M}(\mathcal{E})} \rightarrow \sigma^{\mathcal{M}(\mathcal{E})})$$

は射影。また、 $\llbracket x \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} : X^{\mathcal{M}(\mathcal{E})} \times \sigma^{\mathcal{M}(\mathcal{E})} \rightarrow \sigma^{\mathcal{M}(\mathcal{E})}$ も射影である。
従って、

$$\llbracket x \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1_{X \mathcal{M}(\mathcal{E})}, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})}) = \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})} = \llbracket x[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})}$$

(b) $x \neq y$ なら $\llbracket x[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})} = \llbracket x \rrbracket_X^{\mathcal{M}(\mathcal{E})} : X^{\mathcal{M}(\mathcal{E})} \rightarrow \sigma^{\mathcal{M}(\mathcal{E})}$ は射影である。

この時、射影 $\llbracket x \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} : X^{\mathcal{M}(\mathcal{E})} \times \tau^{\mathcal{M}(\mathcal{E})} \rightarrow \sigma^{\mathcal{M}(\mathcal{E})}$ は合成

$$X^{\mathcal{M}(\mathcal{E})} \times \tau^{\mathcal{M}(\mathcal{E})} \xrightarrow{\text{proj.}} X^{\mathcal{M}(\mathcal{E})} \xrightarrow{\llbracket x \rrbracket_X^{\mathcal{M}(\mathcal{E})}} \sigma^{\mathcal{M}(\mathcal{E})}$$

として書ける。よって

$$\llbracket x \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1_{X^{\mathcal{M}(\mathcal{E})}}, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})}) = \llbracket x \rrbracket_X^{\mathcal{M}(\mathcal{E})} = \llbracket x[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})}$$

2. $U = *$ の時は右辺も左辺も共に一意な射 $X^{\mathcal{M}(\mathcal{E})} \rightarrow 1$ になる。

3. $X, y^v \vdash U : \sigma, X, y^v \vdash V : \tau, X \vdash W : v, y \notin X$ のとき

$$\begin{aligned} & \llbracket (U, V)[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\ &= \llbracket (U[y := W], V[y := W]) \rrbracket_X^{\mathcal{M}(\mathcal{E})} \quad (\text{代入した}) \\ &= (\llbracket U[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})}, \llbracket V[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \quad (\text{解釈の定義}) \\ &= (\llbracket U \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})}), \llbracket V \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})})) \quad (\text{再帰の仮定}) \\ &= (\llbracket U \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})}, \llbracket V \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})}) \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \quad (\text{直積の性質}) \\ &= \llbracket (U, V) \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \quad (\text{解釈の定義}) \end{aligned}$$

4. $X, y^v \vdash U : \sigma \times \tau, X \vdash V : v, y \notin X$ のとき、

$$\begin{aligned} & \llbracket \pi_i(U)[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\ &= \llbracket \pi_i(U[y := V]) \rrbracket_X^{\mathcal{M}(\mathcal{E})} \quad (\text{代入した}) \\ &= \pi_i \circ \llbracket U[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})} \quad (\text{解釈の定義}) \\ &= \pi_i \circ \llbracket U \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \quad (\text{再帰の仮定}) \\ &= \llbracket \pi_i(U) \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \quad (\text{解釈の定義}) \end{aligned}$$

5. 関数記号 $f : \sigma_1, \dots, \sigma_n \rightarrow \tau$ と $X, y^v \vdash U_i : \sigma_i, X \vdash V : v$ に対し

$$\begin{aligned} & \llbracket f((U_i)_{i=1}^n)[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\ &= \llbracket f((U_i[y := V])_{i=1}^n) \rrbracket_X^{\mathcal{M}(\mathcal{E})} \quad \text{代入した} \\ &= f^{\mathcal{M}(\mathcal{E})} \circ (\llbracket U_i[y := V] \rrbracket_X^{\mathcal{M}(\mathcal{E})})_{i=1}^n \quad (\text{解釈の定義}) \\ &= f^{\mathcal{M}(\mathcal{E})} \circ (\llbracket U_i \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})}))_{i=1}^n \quad (\text{再帰の仮定}) \\ &= f^{\mathcal{M}(\mathcal{E})} \circ (\llbracket U_i \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})})_{i=1}^n \circ (1, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \quad (\text{直積の性質}) \\ &= \llbracket f((U_i)_{i=1}^n) \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket V \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \quad \text{解釈の定義} \end{aligned}$$

6. $X, y^v \vdash U : \sigma, X, y^v \vdash V : \sigma, X \vdash W : v, y \notin X$ に対し

$$\llbracket (U =_\sigma V)[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})}$$

$$\begin{aligned}
&= \llbracket (U[y := W] =_{\sigma} V[y := W]) \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\
&= (=_{\sigma \mathcal{M}(\mathcal{E})}) \circ (\llbracket U[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})}, \llbracket V[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \\
&= (=_{\sigma \mathcal{M}(\mathcal{E})}) \circ (\llbracket U \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})}, \llbracket V \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})}) \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \\
&= \llbracket U =_{\sigma} V \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})})
\end{aligned}$$

7. $X, y^v \vdash U : \sigma, X, y^v \vdash V : P(\sigma), X \vdash W : v, y \notin X$ に対し

$$\begin{aligned}
&\llbracket (U \in_{\sigma} V)[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\
&= \llbracket (U[y := W] \in_{\sigma} V[y := W]) \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\
&= (\in_{\sigma \mathcal{M}(\mathcal{E})}) \circ (\llbracket U[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})}, \llbracket V[y := W] \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \\
&= (\in_{\sigma \mathcal{M}(\mathcal{E})}) \circ (\llbracket U \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})}, \llbracket V \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})}) \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \\
&= \llbracket U \in_{\sigma} V \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket W \rrbracket_X^{\mathcal{M}(\mathcal{E})})
\end{aligned}$$

8. $X, x^{\sigma}, y^{\tau} \vdash \varphi : \Omega, X \vdash U : \tau, x \notin X, y \notin X, x \neq y$ として

$$\llbracket \{x : \sigma \mid \varphi\}[y := U] \rrbracket_X^{\mathcal{M}(\mathcal{E})}$$

を考える。

$$\begin{aligned}
&\llbracket \{x : \sigma \mid \varphi\}[y := U] \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\
&= \llbracket \{x : \sigma \mid \varphi[y := U]\} \rrbracket_X^{\mathcal{M}(\mathcal{E})} \\
&\stackrel{1:1}{\rightsquigarrow} \llbracket \varphi[y := U] \rrbracket_{X,x}^{\mathcal{M}(\mathcal{E})} \\
&= \llbracket \varphi \rrbracket_{X,x,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket U \rrbracket_{X,x}^{\mathcal{M}(\mathcal{E})})
\end{aligned}$$

今、 $\llbracket U \rrbracket_{X,x}^{\mathcal{M}(\mathcal{E})} = \llbracket U \rrbracket_X^{\mathcal{M}(\mathcal{E})} \circ \pi_{X \mathcal{M}(\mathcal{E})}$ なので、以下は可換：

$$\begin{array}{ccc}
\text{Hom}_{\mathcal{E}}(X^{\mathcal{M}(\mathcal{E})} \times \sigma^{\mathcal{M}(\mathcal{E})} \times \tau^{\mathcal{M}(\mathcal{E})}, \Omega) & \xrightarrow{\sim} & \text{Hom}_{\mathcal{E}}(X^{\mathcal{M}(\mathcal{E})} \times \tau^{\mathcal{M}(\mathcal{E})}, \Omega) \\
\downarrow - \circ (1, \llbracket U \rrbracket_{X,x}^{\mathcal{M}(\mathcal{E})}) & & \downarrow - \circ (1, \llbracket U \rrbracket_X^{\mathcal{M}(\mathcal{E})}) \\
\text{Hom}_{\mathcal{E}}(X^{\mathcal{M}(\mathcal{E})} \times \sigma^{\mathcal{M}(\mathcal{E})}, \Omega) & \xrightarrow{\sim} & \text{Hom}_{\mathcal{E}}(X^{\mathcal{M}(\mathcal{E})}, \Omega)
\end{array}$$

よって $\llbracket \varphi \rrbracket_{X,x,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket U \rrbracket_{X,x}^{\mathcal{M}(\mathcal{E})})$ は $\llbracket \{x : \sigma \mid \varphi\} \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket U \rrbracket_X^{\mathcal{M}(\mathcal{E})})$ と対応する。

従って、

$$\llbracket \{x: \sigma \mid \varphi\} [y := U] \rrbracket_X^{\mathcal{M}(\mathcal{E})} = \llbracket \{x: \sigma \mid \varphi\} \rrbracket_{X,y}^{\mathcal{M}(\mathcal{E})} \circ (1, \llbracket U \rrbracket_X^{\mathcal{M}(\mathcal{E})})$$



1.2. 健全性定理の証明

2. 完全性定理

2.1. 補題たち

2.2. 完全性定理の証明